Supply Chain Efficiency & Inventory Optimization Dashboard

Project Summary | Power BI Portfolio Project | Synthetic sample dataset

1. Project Objective

Build a single-page Power BI dashboard to monitor inventory levels, stockout and reorder risk, supplier delivery reliability, demand forecast performance, and estimated inventory holding cost. The goal is to turn operational data into clear, actionable insights that support replenishment planning and supply-chain decisions.

2. Dataset & Method

The workbook contains 180 synthetic product/warehouse observations across 2025, with 26 fields covering product, supplier, region, warehouse, stock quantities, lead and delivery times, forecast and actual demand, and estimated holding cost. The Power BI model uses a FactSupplyChain table and a separate DimDate calendar table related by Date.

Key definitions: Closing stock = opening stock + incoming stock – units sold. A stockout observation is flagged when closing stock is zero or actual demand exceeds units sold. Reorder is required when closing stock is at or below reorder level. On-time delivery is evaluated by comparing delivery time with lead time.

3. Key Business Insights

3.1 Inventory risk and replenishment

113 of 180 observations (62.8%) were flagged for stockout risk. Bluetooth Headphones had the highest count of stockout flags (11) and 5 reorder-required observations. Water Bottle had 8 stockout flags, while File Folder Pack had 7. Prioritize reviewing replenishment settings for Bluetooth Headphones and validate whether demand, stock availability, or recording rules explain the flags. Note that a flag does not always mean physical stock was zero.

3.2 Supplier delivery performance

On-time delivery was 72.2% (130 of 180 observations), while 50 observations (27.8%) were marked delayed. PackRight Solutions had the lowest on-time rate at 56.3% (7 delayed observations). OfficeHub Wholesale and PaperTrail Distributors each had 8 delayed observations. HomeEase Trading had the highest on-time rate at 90.5%. Investigate PackRight's delays and compare supplier lead-time commitments, actual delivery times, and delay causes.

3.3 Regional and warehouse patterns

Region	Stockout rate	On-time rate	Units sold
West	57.7%	76.9%	2,002
East	64.0%	72.0%	967
North	63.2%	63.2%	1,480
South	66.2%	73.8%	2,154

South recorded the highest units sold and the highest regional stockout rate, so its replenishment capacity deserves review. North had the lowest regional on-time rate. Mumbai had the lowest warehouse stockout rate (48.4%), while Chennai had the highest (75.0%); Chennai and Pune nevertheless showed high on-time rates (85.0% and 85.7%). This suggests availability and delivery reliability should be investigated separately rather than assuming one explains the other.

3.4 Demand forecast performance

Total forecast demand was **7,045 units**, versus **7,101 units** of actual demand, a net difference of 56 units. However, the weighted absolute percentage error (WAPE) was **15.7%**, with 1,118 units of absolute forecast error. Although aggregate totals are close, errors in individual observations may still affect replenishment decisions. Review forecast misses by product, region, and month before changing planning parameters.

3.5 Inventory movement proxy by category

Product category	Average turnover proxy
Electronics	0.631
Stationery	0.441
Home & Living	0.431
Industrial Supplies	0.342
Sports & Fitness	0.287

Electronics had the highest average turnover proxy and Sports & Fitness the lowest. Use this only as a relative indicator in this sample dataset. It is not a formal financial inventory-turnover measure, which normally requires cost of goods sold and average inventory value over a defined period.

4. Recommended Business Actions

Priority	Recommended action	Reason
High	Review Bluetooth Headphones reorder level and replenishment frequency.	Highest stockout-flag count and repeated reorder alerts.
High	Investigate PackRight Solutions delivery delays.	Lowest observed on-time rate (56.3%).
High	Review South region stock planning and North region delivery reliability.	South has the highest regional stockout rate; North has the lowest on-time rate.
Medium	Analyze forecast errors by product, region, and month.	WAPE is 15.7% despite close aggregate demand totals.
Medium	Review Sports & Fitness inventory movement.	Lowest average turnover proxy in the sample.
Medium	Confirm the currency and time basis for holding-cost values.	Dataset does not establish a verified currency or cost period.

5. Suggested Dashboard Components

The single-page report can include KPI cards for stockout observations, reorder-required observations, on-time delivery percentage, average lead time, WAPE, and estimated holding cost; a category inventory column chart; a supplier delivery performance bar chart; a monthly units-sold trend; a delivery-status donut chart; a forecast-versus-actual demand chart; and a product stock/reorder table. Use slicers for Region, Supplier, Product Category, Warehouse Location, and Date Range.

6. Data Limitations & Interpretation Notes

• The dataset is synthetic and is intended for learning and portfolio demonstration, not real operational decisions.

• Rows represent product/warehouse observations rather than confirmed unique shipment records; delivery metrics are observation-level because no shipment ID is available.

• Holding-cost values have no verified currency or time period, so they should be described as estimated/sample values.

• The turnover ratio is a proxy, not a formal financial inventory-turnover calculation.

• Stockout status uses the rule closing stock = 0 OR actual demand > units sold; it should be described as a stockout-risk flag rather than proof of zero physical inventory.

• Observed closing stock summed across rows is not equivalent to a unique, current inventory balance.

7. Conclusion

The sample analysis highlights inventory availability, supplier reliability, and forecast error as the main areas for investigation. A practical next step is to use the dashboard slicers to examine whether the patterns persist across products, regions, warehouses, and months, then validate the findings against real operational data before making business decisions.